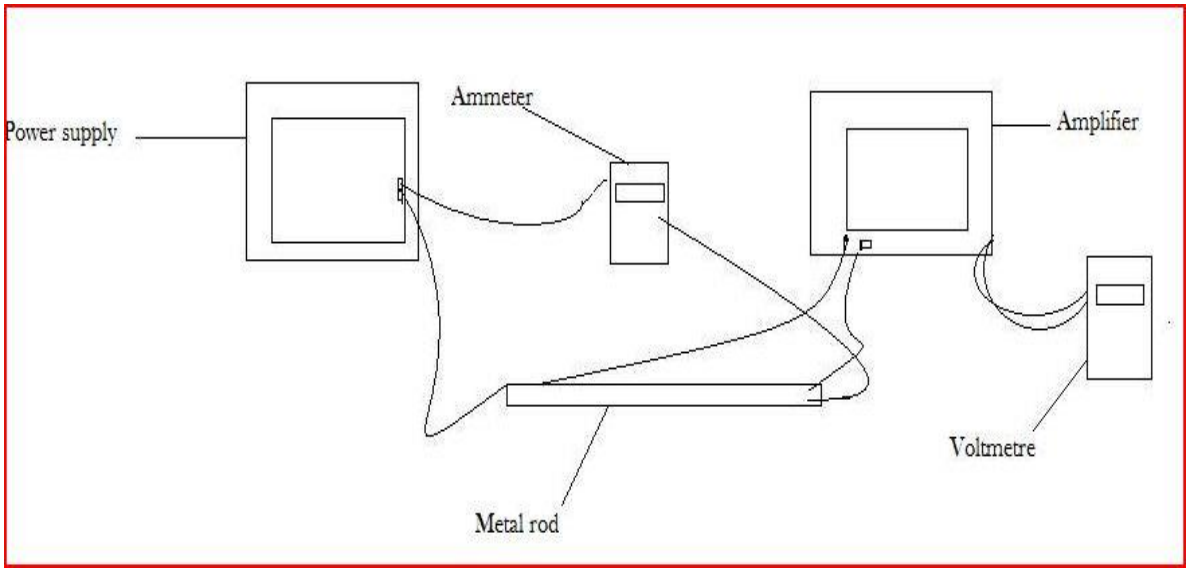
AIM

The aim is to plot the current/voltage characteristics of metal rods (copper and aluminium) and to calculate their resistivity.

APPARATUS

- Heat conductivity rod (copper)
- Heat conductivity rod (aluminium)
- Universal measuring amplifier
- Digital multimeter
- Power supply 0-12V, DC/6V, 12V/AC
- Connecting rod, l = 100mm, yellow
- Connecting rod, l = 250mm, red
- Connecting rod, l = 250mm, blue
- Connecting rod, l = 500mm, red
- Connecting rod, l = 500mm, blue
- Connecting rod, l = 750mm, yellow
- Connecting rod, l = 750mm, blue
- Connecting rod, l = 2000mm, yellow

DIAGRAM



THEORY

The resistivity ρ of a metal is determined from the resistance R of the rod and its dimensions. The rod has a diameter of 2.5cm (cross sectional area $A = \frac{\pi d^2}{4} = 4.91 \times 10^{-4} m^2$) and is 31.5cm long (i.e. length l) between the two voltmeter connections.

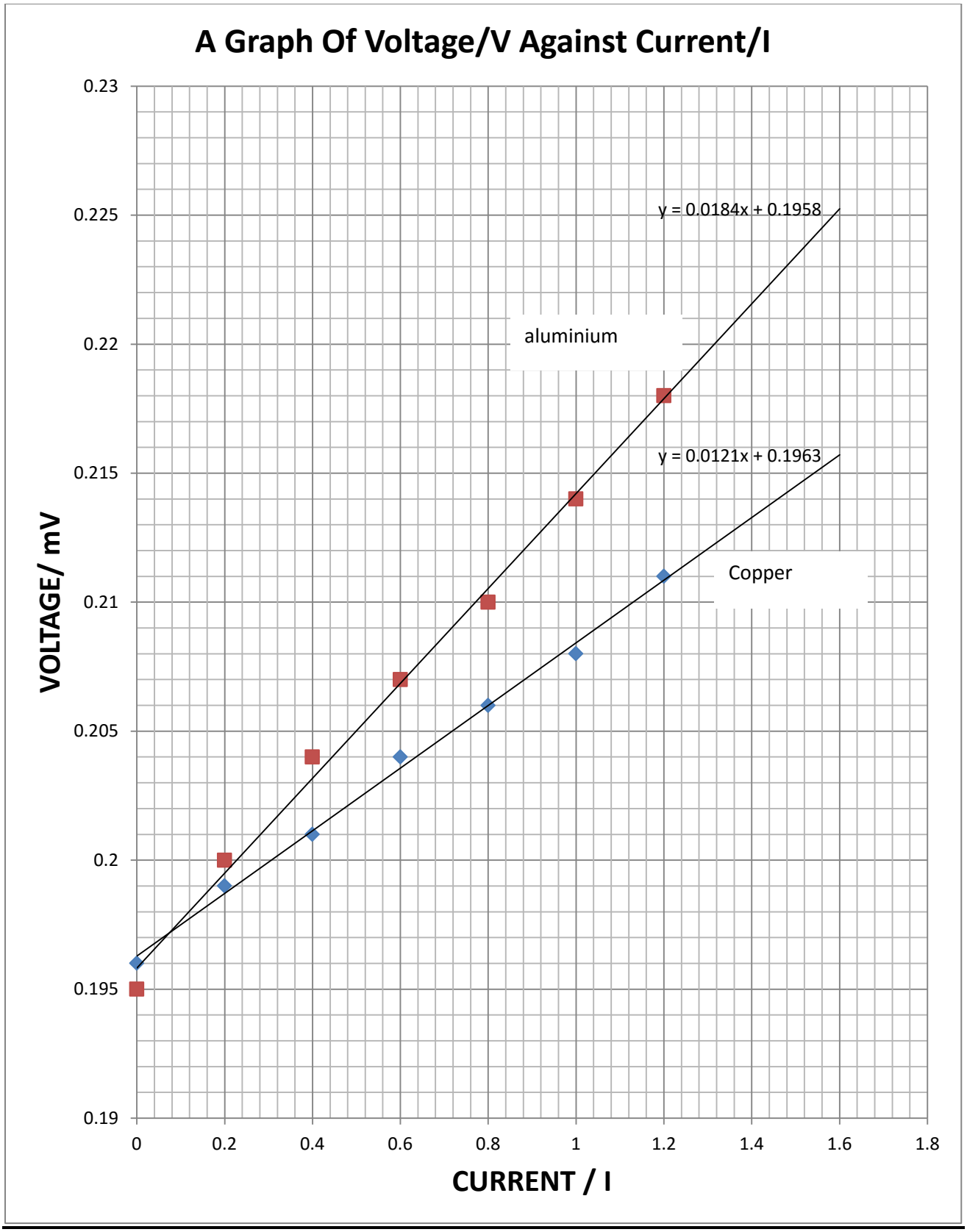
From ohm's law, the resistance of the rod is given as $R \propto \frac{l}{A} \rightarrow R = \rho \frac{l}{A}$

PROCEDURE

The copper rod was connected to the mains with an ammeter as shown in the diagram above. The voltage source was set to 12V. The current was varied at constant intervals of 0.2A. The voltage drop across the rod at the two sockets on the side were measured and recorded using the amplifier for each current selected. The experiment was then repeated using the aluminium rod in place of the copper rod. A graph of voltage against current was plotted for each metal rod on the same graph sheet and the slope was determined. The slope was then used to determine the resistivity of the metal rods.

TABLE OF RESULTS

COPPER			**ALUMINIUM**	
I/A	Pd /mV		I/A	Pd/mV
0.0	0.196		0.0	0.195
0.2	0.199		0.2	0.200
0.4	0.201		0.4	0.204
0.6	0.204		0.6	0.207
0.8	0.206		0.8	0.210
1.0	0.208		1.0	0.214
1.2	0.211		1.2	0.218


A Graph Of Voltage/V Against Current/I
0.23
0.225
0.22
0.215
0.21
0.205
0.2
0.195
0.19
VOLTAGE/ mV
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
CURRENT / I
y = 0.0184x + 0.1958
y = 0.0121x + 0.1963
aluminium
Copper

CALCULATION

For the copper rod

$$slope = \frac{\Delta y}{\Delta x} = \frac{(0.204 - 0.197) \times 10^{-3} V}{(0.64 - 0.06) I} = 1.207 \times 10^{-5} \Omega$$

For the Aluminium rod,

$$slope = \frac{\Delta y}{\Delta x} = \frac{(0.212 - 0.198) \times 10^{-3} V}{(0.88 - 0.12) I} = 1.842 \times 10^{-5} \Omega$$

DISCUSSION

The graph of voltage against current produces a slope which is equivalent to the resistance of the rod used i.e. $V \propto I => V = IR$ where R is the resistance of the rod.

So for the copper rod the slope, and in that case the resistance, was found to be $1.207 \times 10^{-5} \Omega$.

And for the aluminium rod, the slope or resistance was found to be $1.842 \times 10^{-5} \Omega$. The resistance calculated from both rods though are different from the theoretical values of resistances ( ) for copper and aluminium respectively. This deviation could probably be attributed to wrongful conversions and calculation.

SOURCES OF ERROR

Due to the use of the amplifier, there might have been wrong conversion of the values read into the actual values.

PRECAUTIONS

1. The experiment was repeated for each rod so as to produce consistent values on the voltmeter.
2. Whenever the experiment was repeated, the power supply was switched off and the connection cords were inspected again to make sure they were all at their right places.

CONCLUSION

The resistance of the copper and aluminium rods was $1.207 \times 10^{-5} \Omega$ and $1.842 \times 10^{-5} \Omega$ respectively (a deviation from the theoretical values).

REFERENCE

Walter C. MICHELS et al., PHYSICS: Principles and Application; Houghton Mifflin Company, Boston, 1977.